

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 Find each of the following

(a) $\lim_{x \rightarrow 1} -\frac{1}{7} \left(\frac{x^3 - 1}{1 - x} \right).$ [4 marks]

(b) $\lim_{x \rightarrow 3^+} \frac{2|3 - x|}{9 - x^2}$ [5 marks]

2 (a) If $5y^2 + 3xy = 45$, find $\frac{dy}{dx}$. Then, evaluate $\frac{dy}{dx}$ when $x = 0$. [4 marks]

(b) Given that $f(x) = \frac{5}{x^2} + x$. By using the first principle, find $f'(x)$. [5 marks]

3 A curve has the equation $y = \frac{1}{3}x^3 - 2x^2 + 3x + 8$. Find the coordinates of the stationary point and determine whether this is a maximum or a minimum. [7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1 Given that $\sqrt{3+4i} = a+bi$. Find all the possible values of $\sqrt{3+4i}$. [5 marks]

2 Find the solution set for

(a) $\frac{1}{x^2-4} > \frac{1}{5}$ [7 marks]

(b) $7^{1-x} + 7^{x-1} = 2$ [4 marks]

3 Given that $f(ax+b) = \frac{2x+4}{4x+3}$. Find $f(x)$ in terms of a and b .

Hence, if $f(x) = \frac{x+3}{2x+1}$, obtain the values of a and b . [10 marks]

4 (a) Show that the function $f(x) = 1 - \sqrt{2-x}$ has an inverse. State the domain and the range for $f(x)$. [5 marks]

(b) Given $g(x) = \begin{cases} |x| & , -2 \leq x < 0 \\ 5 & , x = 0 \\ x^3 & , x > 0 \end{cases}$

(i) State the domain of $g(x)$ [1 marks]

(ii) Sketch the graph of $g(x)$. Hence, find the range of $g(x)$ [4 marks]

5 The function $f(x)$ is defined by $f(x) = \begin{cases} x-4 & , x < 2 \\ A & , x = 2 \\ 2x+B & , 2 < x \leq 5 \\ C & , x > 5 \end{cases}$ where A , B and C are

constants.

(a) Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$. [4 marks]

(b) Use the definition of continuity to determine the values of A and B if $f(x)$ is continuous at $x = 2$. [6 marks]

- (c) Show that if $\lim_{x \rightarrow 5} f(x)$ does not exist, then $C \neq 4$. Is $f(x)$ continuous at $x = 5$ when $C \neq 4$. [4 marks]

6 Given $e^{-2y} + 2xy + \ln(1 - 2x) = 2$.

- (a) Calculate the value of y when $x = 0$. [3 marks]

- (b) By differentiating with respect to x implicitly, show that

$$\left(x - e^{-2y}\right) \frac{d^2 y}{dx^2} + 2e^{-2y} \left(\frac{dy}{dx}\right)^2 + 2 \frac{dy}{dx} - \frac{2}{(1 - 2x)^2} = 0. \quad [9 \text{ marks}]$$

- (c) Use the result obtain from (a) and (b) to find the values of $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$. Give the answer correct to two decimal places. [4 marks]

7 A cuboid has a rectangular base of length $2x$ cm, width x cm and height y cm. If the volume of the cuboid is 72cm^3

- (a) Show the surface area A cm², is given by

$$A = 4x^2 + \frac{216}{x}$$

- (b) Determine the value x for which the surface area of the cuboid is minimum. Hence, find the minimum surface area of the cuboid.

[9 marks]

END OF QUESTION PAPER

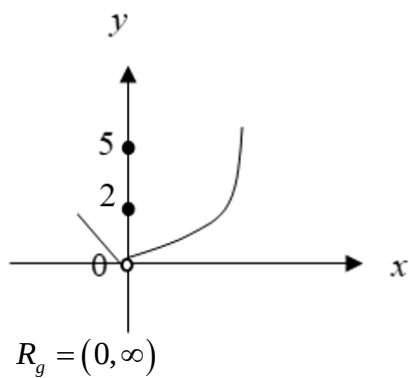
Answers

Section A

- 1 (a) $\frac{3}{7}$ (b) $-\frac{1}{3}$
- 2 (a) $-\frac{3}{10}$ (b) $-\frac{10}{x} + 1$
- 3 Maximum point $\left(1, \frac{28}{3}\right)$, Minimum point $(3, 8)$.

Section B

- 1 $\sqrt{3+4i} = \pm(2+i)$
- 2 (a) $\{x: -3 < x < -2 \text{ or } 2 < x < 3\}$ (b) $\{x: x=1\}$
3. $a=2$ $b=1$
- 4 (a) $D_f: (-\infty, 2], R_f: (-\infty, 1]$
 (i) $D_g: [-2, \infty)$
 (ii)



- 5 (a) $\lim_{x \rightarrow 2^-} f(x) = -2, \lim_{x \rightarrow 2^+} f(x) = 4 + B$
 (b) $A = -2, B = -6$
 (c) shown, $f(x)$ is not continuous
- 6 (a) $y = -\frac{1}{2} \ln 2$
 (b) shown
 (c) $\frac{dy}{dx} = -0.67, \frac{d^2y}{dx^2} = -0.77$
- 7 (a) shown
 (b) $x = 3, A = 108 \text{ cm}^3$